ModelIt!

Student Activity Pack

Building for the Big One: Earthquake Engineering Lab

How Engineers Design Buildings That Bend Instead of Break

Grade 04L2 | NGSS 4-ESS3-2, 4-ETS1-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think some buildings fall down during earthquakes while others stay standing?

2 Would you rather be in a building made of rigid concrete or one that can sway back and forth during an earthquake? Why?

3 Draw what you think happens to a building when the ground underneath it starts shaking.

4 What do you think engineers do differently when they build buildings in earthquake zones?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Earthquake Magnitude | Building Flexibility | Shaking Force | Structural Damage | Occupant Safety

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Small Earthquake, Rigid Building

Earthquake Magnitude at 30%, Building Flexibility at 20%

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Big Earthquake, Rigid Building

Earthquake Magnitude at 90%, Building Flexibility at 20%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Big Earthquake, Flexible Building

Earthquake Magnitude at 90%, Building Flexibility at 80%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Foundation Type — How the building connects to the ground — some foundations are rigid while others use base isolation that allows the building to slide independently of the ground

Building Height — How tall the building is — taller buildings experience more swaying at the top during earthquakes

Material Strength — How strong the building materials are — reinforced concrete and steel are stronger than brick and unreinforced masonry

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do some buildings collapse during earthquakes while others right next to them survive with barely a crack? Use evidence from your simulation to support your answers.

1. **Why is a flexible building safer than a rigid one during an earthquake?**

2. **How is a building swaying in an earthquake similar to a tree bending in a strong wind?**

3. **If you were designing a hospital in an earthquake zone, what would be your top three design priorities?**

4. **Why can we not prevent earthquakes, so we have to design buildings that survive them?**

5. **How did your model help you understand what makes the difference between buildings that survive and buildings that collapse?**

8 STEM Challenge: Earthquake-Proof Tower Challenge

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YOUR ENGINEERING MISSION

Design and build the tallest tower that can survive a simulated earthquake (shaking table) without collapsing.

THE SCENARIO

A city in an earthquake zone needs a new hospital that can survive a major earthquake and keep patients safe. Your engineering team must design and build a tower model that is as tall as possible while surviving a simulated earthquake on a shaking table. Use your model evidence to decide what materials and shapes make the most earthquake-resistant structure.

Engineering Design Thinking

1. WHAT BUILDING MATERIALS WILL YOU USE (STRAWS, PIPE CLEANERS, POPSICLE STICKS, MARSHMALLOWS, TAPE)?

2. HOW WILL YOU MAKE YOUR TOWER FLEXIBLE ENOUGH TO SWAY WITHOUT COLLAPSING?

3. WHAT IS THE TALLEST YOU CAN BUILD WHILE STILL SURVIVING THE SHAKE TEST?

4. HOW WILL YOU SIMULATE AN EARTHQUAKE TO TEST YOUR DESIGN (SHAKING A COOKIE SHEET, TAPPING A TABLE)?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Structural Engineers and Seismic Engineers design buildings, bridges, and infrastructure that can withstand earthquakes, hurricanes, and other natural disasters. Their work directly saves thousands of

lives. They earn $70,000–$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 During an earthquake, a flexible building sways back and forth but does not collapse, while a rigid building next to it cracks and crumbles. Why does the flexible building survive?

- **A.** The flexible building is lighter, so the earthquake cannot reach it
- **B.** The flexible building moves WITH the earthquake, absorbing and distributing the shaking energy instead of fighting against it
- **C.** The flexible building is built with stronger materials that cannot be damaged
- **D.** The earthquake was weaker on the side where the flexible building stood

2 In the model, Earthquake Magnitude and Building Flexibility are external components. Why?

- **A.** Because they are the most dangerous parts of the model
- **B.** Because they are the starting conditions, one natural and one engineered, that determine what happens to the internal components
- **C.** Because they happen outside the building
- **D.** Because they cannot be measured by scientists

3 The model shows that Building Flexibility has a NEGATIVE relationship with Structural Damage. What does this mean for earthquake engineering?

- **A.** More flexible buildings cause more damage during earthquakes
- **B.** Building Flexibility and Structural Damage always increase together
- **C.** When Building Flexibility increases, Structural Damage decreases because the building absorbs shaking energy instead of cracking
- **D.** Flexibility has no effect on how much damage a building takes

4 A student compares two scenarios: a magnitude 90% earthquake hitting a building with 20% flexibility versus the same earthquake hitting a building with 80% flexibility. What difference would the model show?

- **A.** Both buildings would have the same damage because the earthquake is the same strength
- **B.** The rigid building (20% flexibility) would have much more Structural Damage and much lower Occupant Safety than the flexible building (80% flexibility)
- **C.** The flexible building would have more damage because bending weakens the structure
- **D.** Neither building would be damaged because modern buildings are always safe

5 **The model shows that Structural Damage has a NEGATIVE relationship with Occupant Safety. Why is this the most important relationship for earthquake engineers?**

- **A.** Because engineers want to increase both damage and safety at the same time
- **B.** Because when Structural Damage increases, Occupant Safety decreases, meaning cracking walls and falling debris put people in danger
- **C.** Because Occupant Safety causes Structural Damage, not the other way around
- **D.** Because this relationship only matters for small earthquakes